Microsoft

Microsoft Developer Experience
GitHub Copilot Demo

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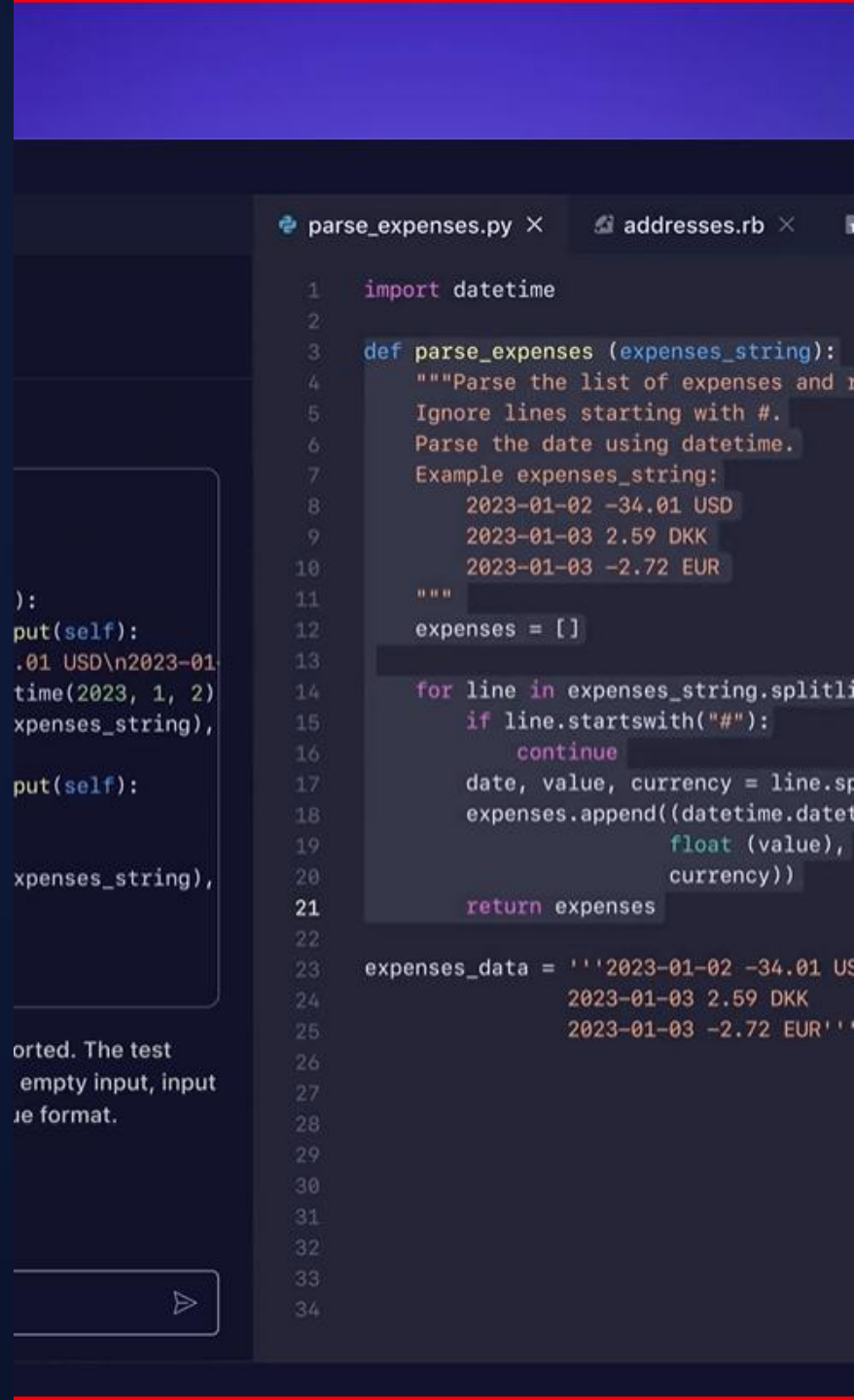
Microsoft

GitHub Copilot

Visual Studio Code

AGENDA

- Prolog:
  - Overview
  - Prompt Engineering
  - Developer copilot interaction
  - Data flows & patterns
  - Azure Functions scenario review
- Demo: GitHub CoPilot scenario(s), usage
- References & Next Steps

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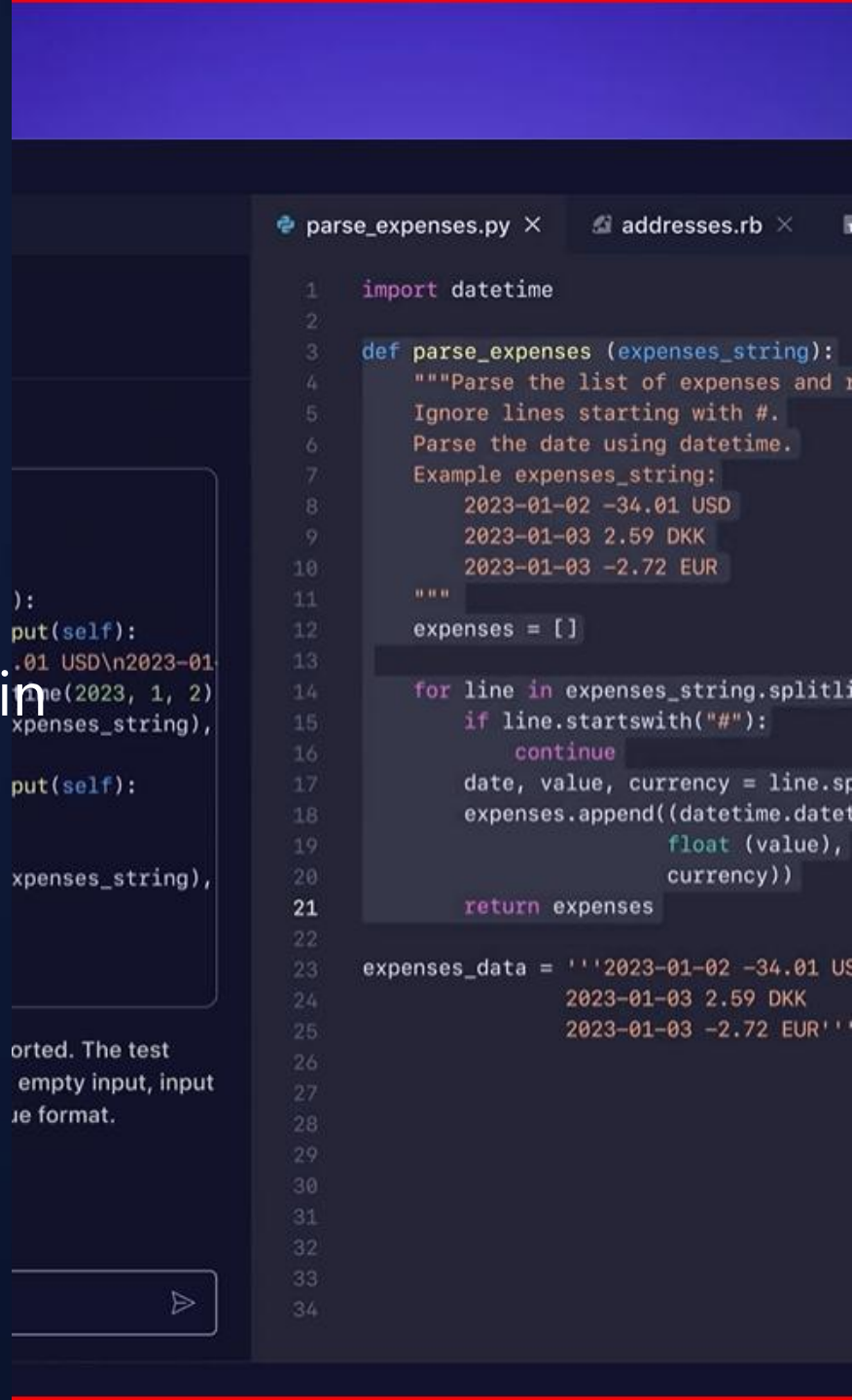
AI pair/peer-programmer

- AI developer tool, helps write code faster with less work
- Draws context from comments & code to suggest code/functions
- Developers code faster, focus on solving problems, stay in context & feel more fulfilled with their work
- Powered by OpenAI Codex, generative pretrained language models



Supported IDEs

Visual Studio

Visual Studio Code

Vim/Neovim

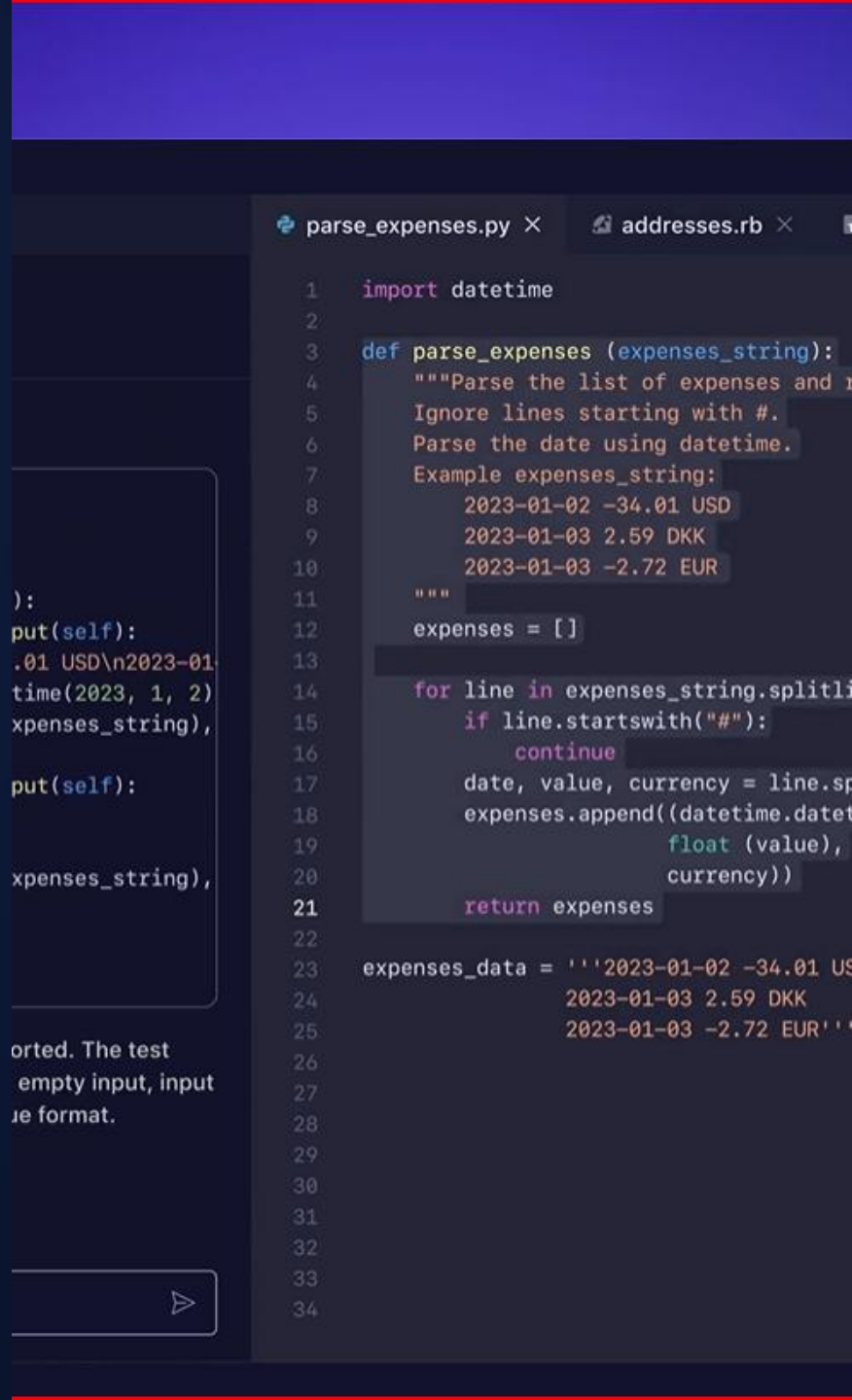
Jet Brains

GitHub CoPilot Emphasis Items

- CoPilot is exactly that, co-pilot, not a replacement
- Training data, hints & suggestions
  - IOW: sometimes, recommendations are wrong
- Predictive, based on semantics & context

Developer is PIC

- Design patterns & structure
- Coding standards, source management & style
- Context & snippets must be vetted for expected output



Limitations

“I’m powered by AI, so surprises and mistakes are possible. Make sure to verify any generated code or suggestions, and share feedback so that we can learn and improve.”

Copilot 

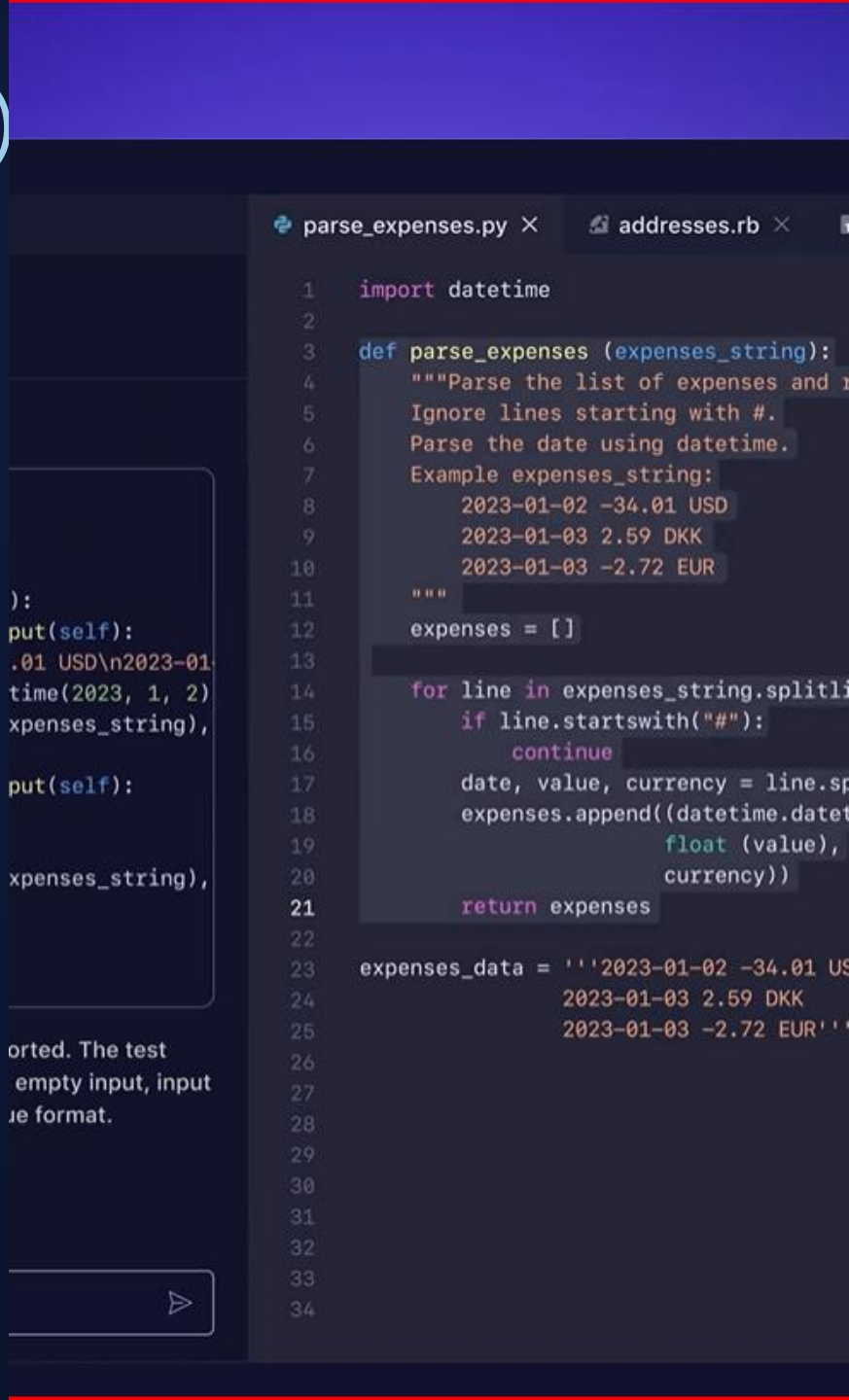
GitHub CoPilot prompt engineering (4s')

- **Single**: focus prompt on a single, well-defined task or question. Clarity is crucial for eliciting accurate and useful responses.

- **Specific**: instructions are explicit and detailed. Specificity leads to more applicable and precise code suggestions.

- **Short**: keep prompts concise and to the point, balance ensures clarity without overloading.

- **Surround**: retain environment elements for tailored code suggestions, filenames, variables, methods



Techniques

Zero-Shot Prompting

No Example

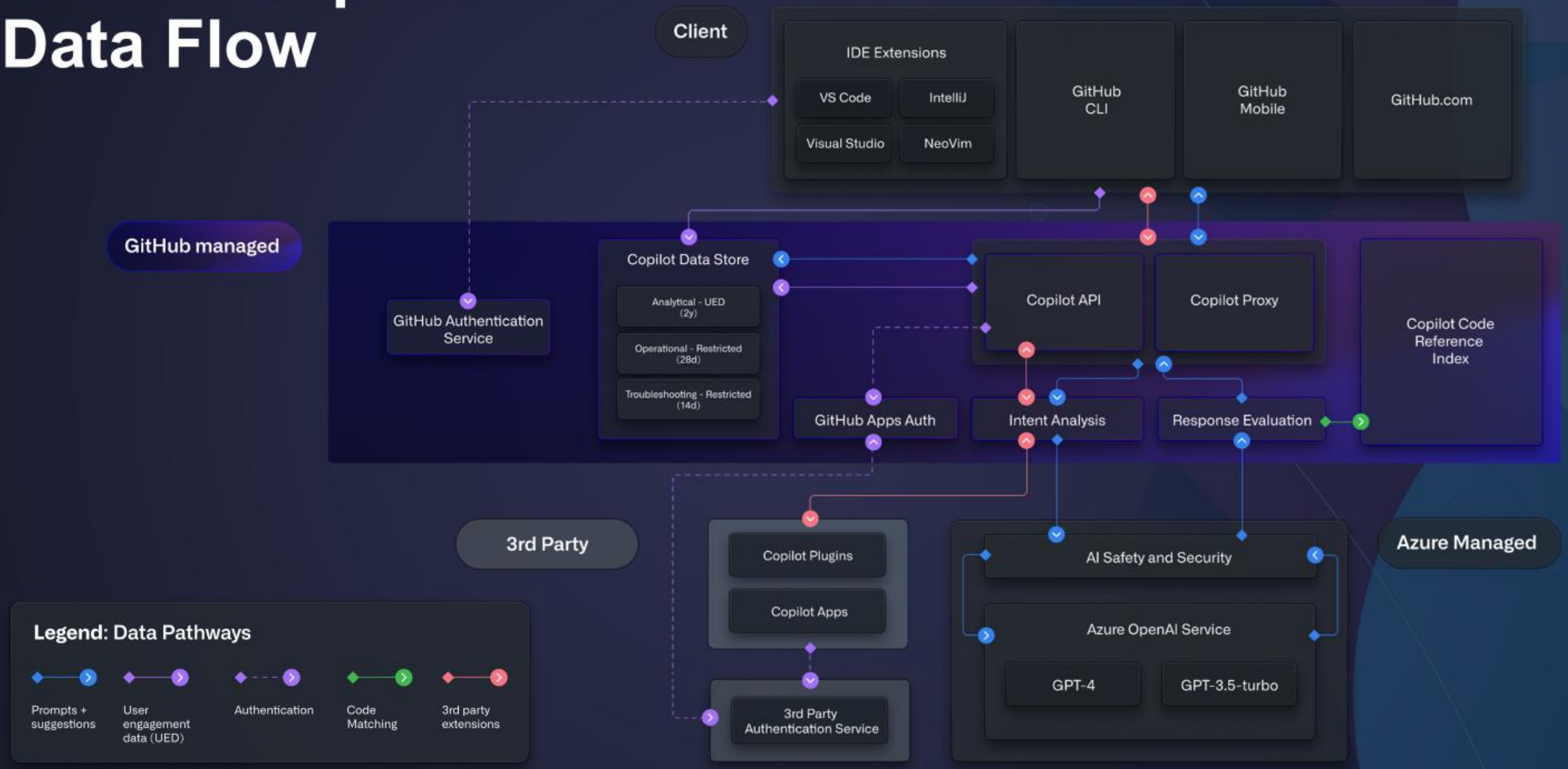
One-Shot Prompting

One Example

Few-Shot Prompting

Handful of Examples

GitHub Copilot Data Flow



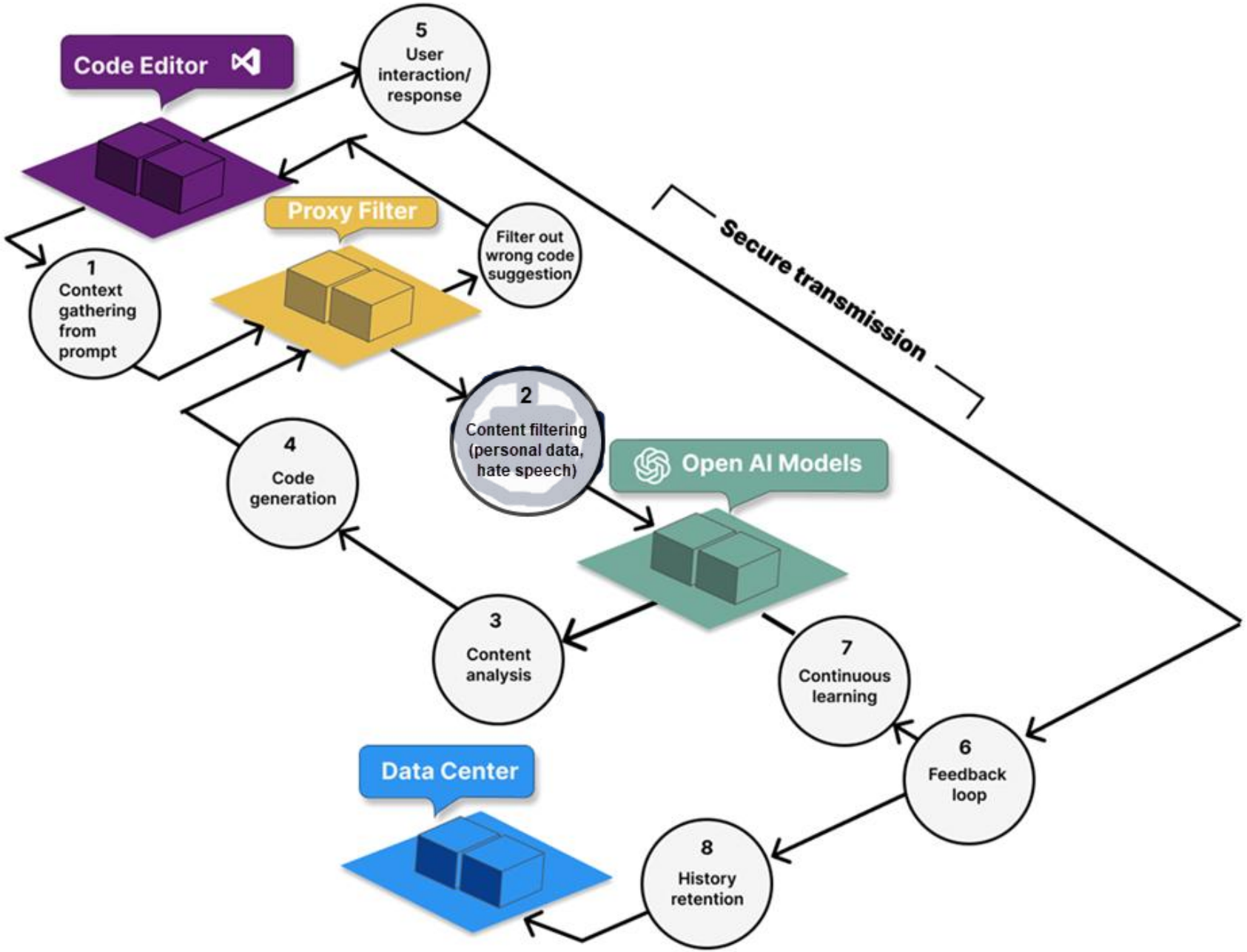


Illustration of Copilot Prompt Processing Flow

GitHub CoPilot references

GitHub CoPilot landing page

GitHub CoPilot Trust Center

GitHub CoPilot LinkedIn Tips & Tricks

GitHub CoPilot for Skeptics, MAY 2025 BRK124

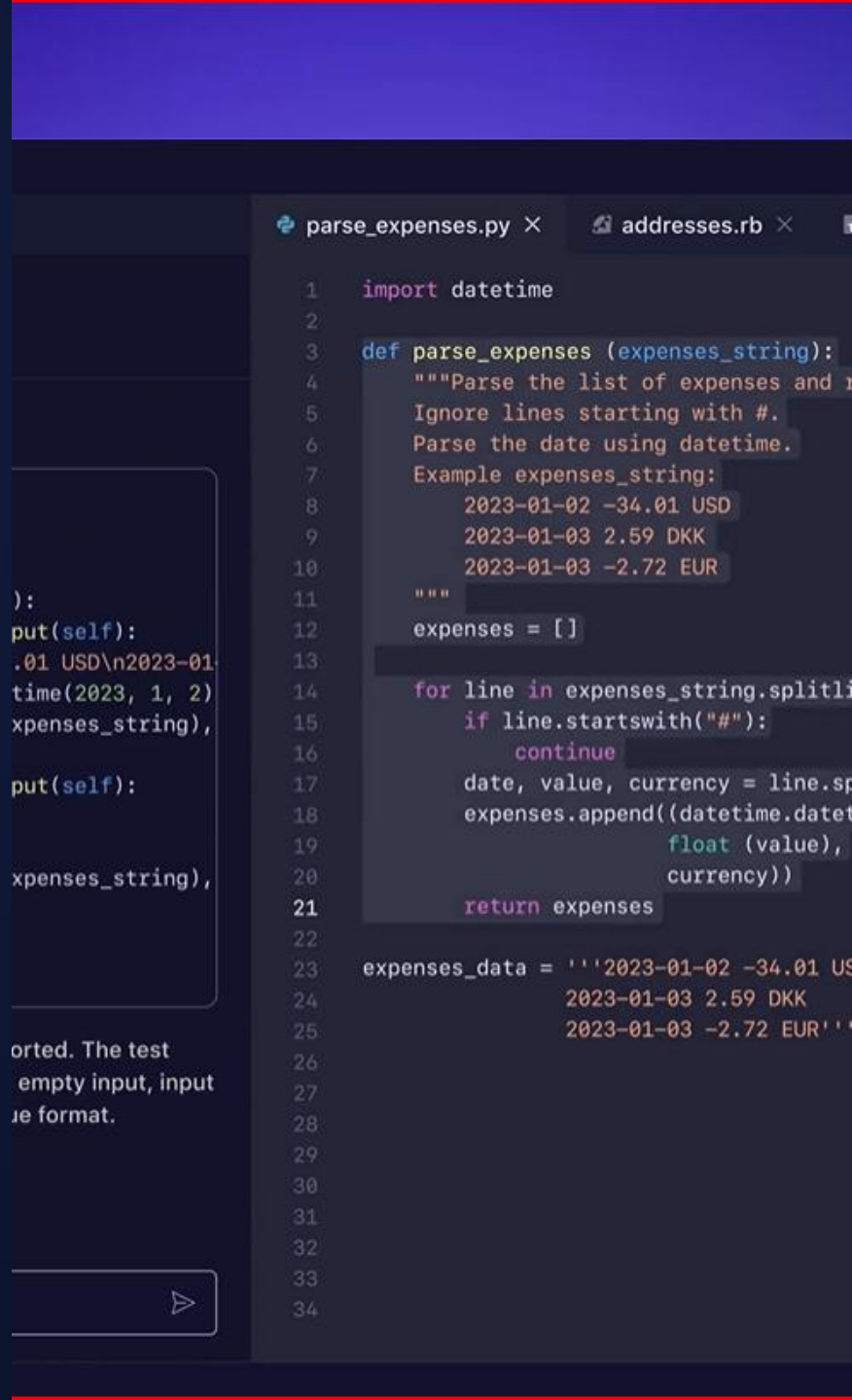
GitHub CoPilot MSFT Learning Path

GitHub CoPilot YouTube Series

GitHub CoPilot Agent Mode 101

Awesome GitHub CoPilot Customizations

Visual Studio Code YouTube channel



Microsoft

GitHub Copilot

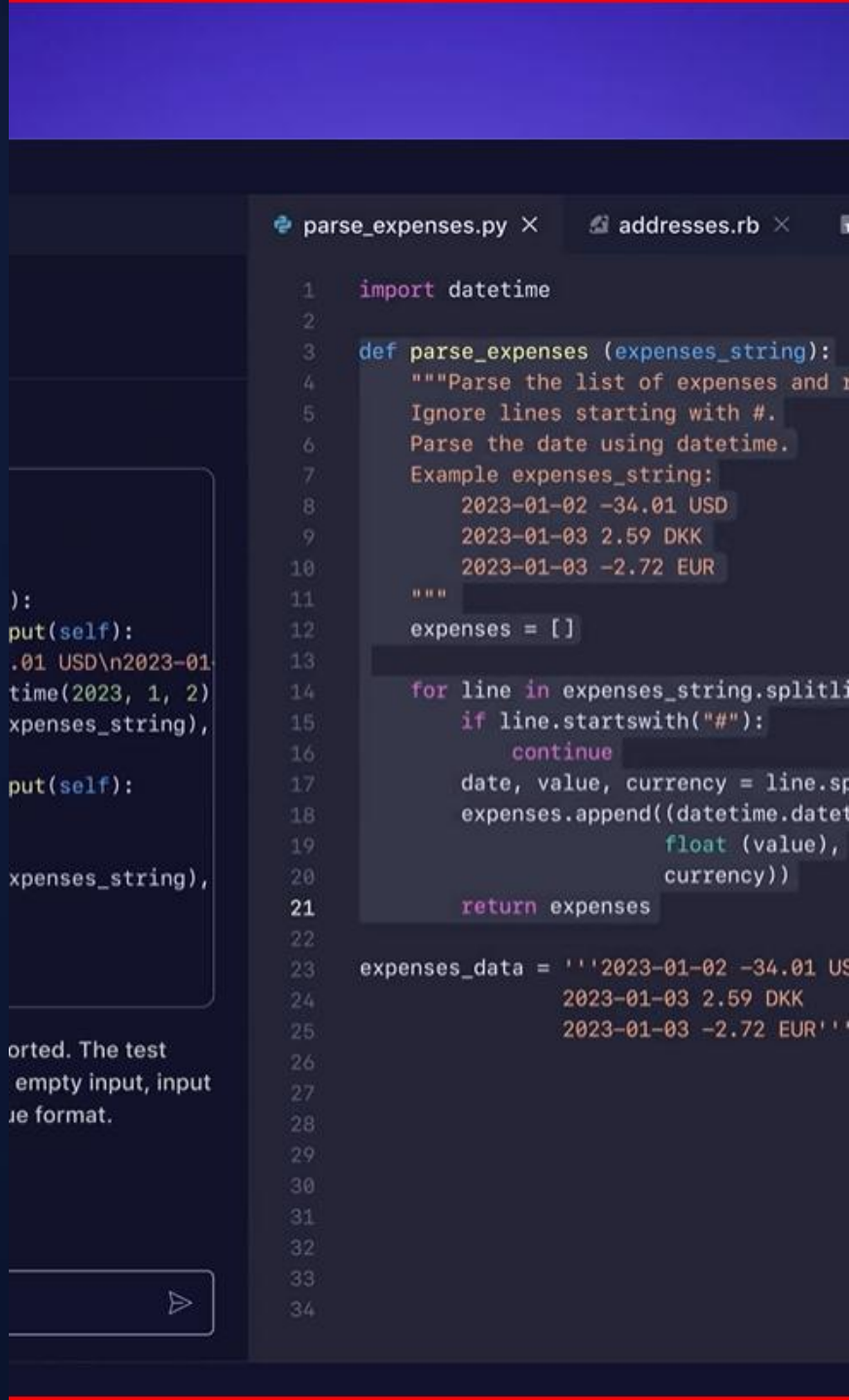
Visual Studio Code

Exploring GitHub Copilot with Microsoft Fabric

- Integrate Fabric Workspaces with a Git repository
  - ✓ Clone the repository & work in Visual Studio Code
  - ✓ GitHub Copilot Scenario
  - ✓ Fabric references
  - ✓ Code changes & commit to Git repo

- Scenario
  - ✓ Developing or extending Fabric items (Data Warehouses, Notebooks, Semantic Models, KQL Databases, etc.)
  - ✓ Code migration
  - ✓ Code refactoring
  - ✓ Readme.md

- Reference scenarios
  - ✓ Fabric data engineering, Spark notebooks
  - ✓ SQL database, Fabric data warehouses & Fabric SQL databases
  - ✓ Tabular model definition, for Fabric semantic models



GitHub Copilot at a glance

AI that builds with you

GitHub Copilot is the world's most widely adopted AI developer tool and the competitive advantage developers ask for by name. It provides AI Assistance throughout the development lifecycle—within the IDE and GitHub.com—to streamline development during planning, coding, pull requests, and more.

Focus on tasks you love—leave the rest to agents

Automate repetitive work, streamline code reviews, and resolve issues with agent mode in integrated development environments (IDEs). Agent mode reasons through the problem, coordinates next steps, and applies the changes—while keeping you in the driver's seat.

GitHub Copilot Business
$19 user / month

Includes 300 premium requests per user month

GitHub Copilot Enterprise
$39 user / month

Includes 1,000 premium requests per user per month

Unlimited use of base model
Additional premium requests at $0.04 USD*

CAPABILITIES	DESCRIPTION
IDE Support	GitHub Copilot supports the most IDEs and integrated with GitHub.com. IDEs supported include Visual Studio Code, Visual Studio, JetBrains, IntelliJ, Eclipse, Neovim, and Xcode.
Inline Code Completions	New custom-trained GPT-4o model and advanced predictive edits will reduce latency.
Agentic AI Development	**Agent Mode** helps you perform sweeping changes quickly by analyzing code, proposing edits, running tests, and validating results across multiple files. **Code review** analyzes your work, uncovers hidden bugs, fixes mistakes, and more—before a human ever sees it.
Multi-Model	Supports latest models from OpenAI, Anthropic, and Gemini. See GitHub docs for model use cases.
Responsible AI, Certifications, Copyright/ Indemnity and Data Privacy	• Enterprise-level control to exclude code and public code license filtering/annotations. See GitHub Copilot Trust Center for details. • SOC 2 Type II, ISO/IEC 27001:2013 certifications and CSA STAR compliance. • Code generated by Copilot is your intellectual property and protected should a copyright issue arise.
Extensibility and Scalability	• Copilot's capabilities can be extended and re-used using VS Code extension APIs • Support for Model Context Protocol (MCP) available • Single sign-on (SSO) in both Business and Enterprise

Capabilities as of April 2025. See the latest announcements at the GitHub Copilot Changelog or reach out to your representative.
*See base and premium model details in GitHub Documentation. Multipliers may apply for select premium models.

GitHub Copilot Scenario

1

Feature pulled from backlog for Sprint

2

Peer development team lacks required **time** & **experience** to implement feature

3

Director requested assistance to implement

4

Complete within 3 days & handoff to team for integration

Feature requirements

1

Develop a compute service endpoint in Azure?

2

Requires a secure method to store secrets?

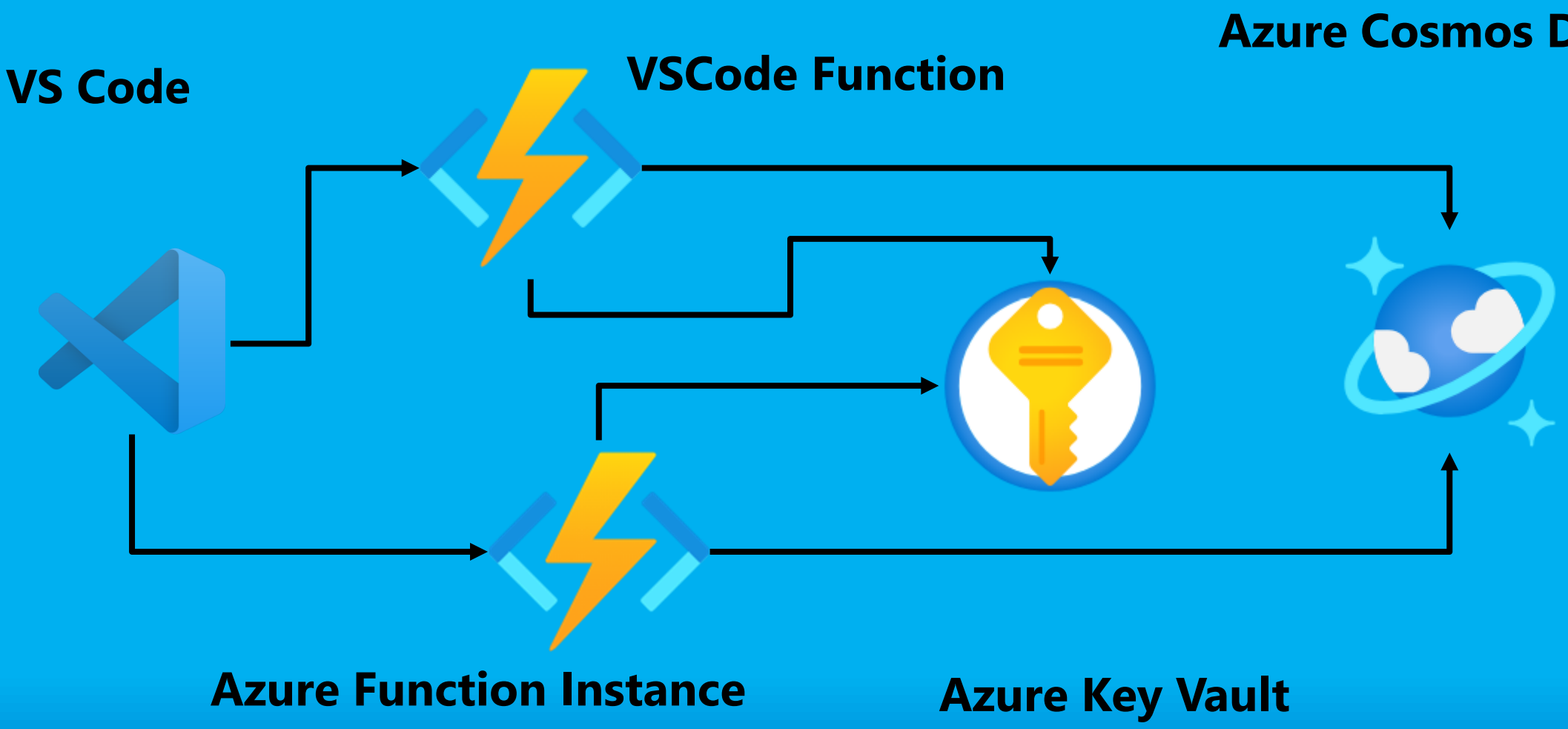
3

Persist data to Cosmos DB?

4

Service must be scalable to handle dynamic workloads?

Conceptual Architecture Diagram



Quick start questions, considerations

- Azure Service: Azure compute decision tree?
- Visual Studio Code Extensions?
- .NET packages required?
- Prototype, scaling, REST endpoint?
- Securely manage, access a secret?
- Store persistent data?
- Challenges
  - ✓ Learning curve & .NET language
  - ✓ KeyVault URI & Secret
    - ➢ secret squirrel nut stash
  - ✓ CosmosDB & Container name

Virtual Joel

Let's fail fast, VS Code & CoPilot

TRADITIONAL APPROACH

- **Microsoft Learn Modules: ~15 hours**
- **Quick Start Modules: 3-4 hours**
- **Stackoverflow**
- **W3Schools, etc ….**
- **Training: internal, external, Udemy, LinkedIn**